Information Theory Mn908 and MICAS

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Model Answers to Exercise 2 of September 30, 2026

Problem 1: Huffman Codes

The first two codes $\mathcal{C}_1$ and $\mathcal{C}_2$ cannot be Huffman codes.

The code $\mathcal{C}_1$ is not a Huffman code because it is not prefix-free. For example, the codeword 10 is a prefix of both 101 and 1010.

The code $\mathcal{C}_2$ is not a Huffman code because it satisfies Kraft's inequality with strict inequality. Indeed,

$$2^{-2} + 4 \cdot 2^{-3} + 3 \cdot 2^{-4} = \frac{1}{4} + \frac{1}{2} + \frac{3}{16} = \frac{15}{16} < 1.$$

Since the Kraft sum is strictly smaller than 1, the code cannot be optimal among binary prefix-free codes: at least one codeword can be shortened while preserving prefix-freeness. Hence it cannot be a Huffman code.

The code $\mathcal{C}_3$ is a Huffman code for a source X with probabilities

$$(p_1, \ldots, p_8) = \left(\frac{1}{4}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{8}, \frac{1}{16}, \frac{1}{16}\right).$$

For these probabilities, a Huffman construction can merge the two symbols of probability 1/16, and the resulting tree has codeword lengths

$$2, 3, 3, 3, 3, 3, 4, 4,$$

which are precisely the lengths of the codewords in $\mathcal{C}_3$.

Problem 2: Optimal Code Lengths that Require Almost One Bit above Entropy

Let X be a binary random variable with probability mass function

$$P_X(x) = \begin{cases} 1-\delta, & x = x_0, \\ \delta, & x = x_1, \end{cases} \qquad 0 < \delta < 1.$$

An optimal binary code assigns to both symbols a codeword of length 1, for example

$$c(x_0) = 0, \qquad c(x_1) = 1.$$

Hence the expected length of this optimal code is

$$L^* = 1.$$

The entropy of X is

$$H(X) = h_b(\delta),$$

where $h_b(\cdot)$ denotes the binary entropy function. Since $h_b(\cdot)$ is continuous and

$$\inf_{\delta \in (0,1)} h_b(\delta) = 0,$$

for every $\varepsilon > 0$ there exists a $\delta \in (0, 1)$ such that $h_b(\delta) < \varepsilon$. Thus

$$L^* = 1 > 1 + h_b(\delta) - \varepsilon = H(X) + 1 - \varepsilon.$$

Problem 3: Codes with Costs

1. We wish to minimize
$$C = \sum_{i=1}^{m} p_i c_i l_i$$
subject to
$$\sum_{i=1}^{m} 2^{-l_i} \leq 1.$$
Define
$$r_i = \frac{2^{-l_i}}{\sum_{j=1}^{m} 2^{-l_j}},$$
and hence
$$l_i = \log_2 \left(\frac{1}{r_i \sum_{j=1}^{m} 2^{-l_j}} \right).$$
Further introduce
$$Q = \sum_{j=1}^{m} p_j c_j, \qquad q_i = \frac{p_i c_i}{Q}.$$
Then $q = (q_1, \ldots, q_m)$ and $r = (r_1, \ldots, r_m)$ are probability distributions, and
$$\begin{aligned} C &= \sum_{i=1}^{m} p_i c_i l_i \\ &= Q \sum_{i=1}^{m} q_i \log_2 \left(\frac{1}{r_i \sum_{j=1}^{m} 2^{-l_j}} \right) \\ &= Q \left(\sum_{i=1}^{m} q_i \log_2 \frac{q_i}{r_i} - \sum_{i=1}^{m} q_i \log_2 q_i - \log_2 \left(\sum_{j=1}^{m} 2^{-l_j} \right) \right) \\ &\geq Q(D(q\|r) + H(q)) \\ &\geq QH(q). \end{aligned}$$
Here, equality holds if and only if
$$\sum_{j=1}^{m} 2^{-l_j} = 1 \qquad \text{and} \qquad r = q.$$
These conditions can simultaneously be satisfied by choosing
$$l_i^* = \log_2 \frac{1}{q_i} = -\log_2 \frac{p_i c_i}{\sum_{j=1}^{m} p_j c_j},$$
where the integer constraints on l_i are ignored. Therefore,
$$C^* = QH(q).$$

2. If we use q instead of p in the Huffman procedure, we obtain a code minimizing the expected costs. Equivalently, we may use the weights $p_i c_i$, since multiplying all weights by the same positive constant does not change the Huffman construction.

3. Since C^* is the minimum cost when the integer constraints on l_i are relaxed,

$$C^* \leq C_{\text{Huffman}}.$$

To upper-bound C_{Huffman}, choose

$$l_i = \lceil -\log_2 q_i \rceil = \left\lceil \log_2 \frac{1}{q_i} \right\rceil.$$

Then

$$\begin{aligned} C &= \sum_{i=1}^{m} p_i c_i l_i \\ &= \sum_{i=1}^{m} p_i c_i \left\lceil \log_2 \frac{1}{q_i} \right\rceil \\ &< \sum_{i=1}^{m} p_i c_i \left(\log_2 \frac{1}{q_i} + 1 \right) \\ &= Q \sum_{i=1}^{m} q_i \log_2 \frac{1}{q_i} + \sum_{i=1}^{m} p_i c_i \\ &= QH(q) + Q \\ &= C^* + Q. \end{aligned}$$

Since Huffman coding is optimal among binary prefix-free codes,

$$C^* \leq C_{\text{Huffman}} < C^* + Q = C^* + \sum_{i=1}^{m} p_i c_i.$$

Problem 4: Entropy Rate

1. Recall that a stochastic process $\{X_i\}_{i \geq 1}$ is stationary if, for all $l, m, k \geq 1$, the joint distribution of

$$(X_l, X_{l+1}, \ldots, X_{l+k})$$

is the same as that of

$$(X_m, X_{m+1}, \ldots, X_{m+k}).$$

For $\{Y_i\}$, we have

$$\begin{aligned} &\Pr[Y_l = y_l, Y_{l+1} = y_{l+1}, \ldots, Y_{l+k} = y_{l+k}] \\ &\quad = \Pr[X_l = x_l, X_{l+1} = x_{l+1}, \ldots, X_{l+k+1} = x_{l+k+1}] \\ &\quad = \Pr[X_m = x_m, X_{m+1} = x_{m+1}, \ldots, X_{m+k+1} = x_{m+k+1}] \\ &\quad = \Pr[Y_m = y_m, Y_{m+1} = y_{m+1}, \ldots, Y_{m+k} = y_{m+k}], \end{aligned}$$

where stationarity of $\{X_i\}$ gives the second equality.

The stationarity of $\{Z_i\}$ and $\{V_i\}$ follows analogously. In particular, stationarity of $\{X_i\}$ implies that the joint distribution of

$$(X_{2l}, X_{2(l+1)}, \ldots, X_{2(l+k)})$$

is independent of the time shift l, which gives stationarity of $\{V_i\}$. Thus all three processes $\{Y_i\}$, $\{Z_i\}$, and $\{V_i\}$ are stationary.

2. For a stationary process $\{X_i\}$, the entropy rate exists and can be written as

$$H(\{X_i\}) = \lim_{n\to\infty} H(X_n \mid X_1, \ldots, X_{n-1}).$$

For $\{Y_i\}$,

$$\begin{aligned} H(\{Y_i\}) &= \lim_{n\to\infty} H(Y_n \mid Y_1, \ldots, Y_{n-1}) \\ &= \lim_{n\to\infty} H(X_n, X_{n+1} \mid X_1, \ldots, X_n) \\ &= \lim_{n\to\infty} H(X_{n+1} \mid X_1, \ldots, X_n) \\ &= H(\{X_i\}). \end{aligned}$$

For $\{Z_i\}$,

$$\begin{aligned} H(\{Z_i\}) &= \lim_{n\to\infty} H(Z_n \mid Z_1, \ldots, Z_{n-1}) \\ &= \lim_{n\to\infty} H(X_{2n}, X_{2n+1} \mid X_1, \ldots, X_{2n-1}) \\ &= \lim_{n\to\infty} H(X_{2n} \mid X_1, \ldots, X_{2n-1}) \\ &\quad + \lim_{n\to\infty} H(X_{2n+1} \mid X_1, \ldots, X_{2n}) \\ &= 2H(\{X_i\}). \end{aligned}$$

Finally,

$$\begin{aligned} H(\{V_i\}) &= \lim_{n\to\infty} H(V_n \mid V_1, \ldots, V_{n-1}) \\ &= \lim_{n\to\infty} H(X_{2n} \mid X_2, X_4, \ldots, X_{2(n-1)}) \\ &\geq \lim_{n\to\infty} H(X_{2n} \mid X_1, X_2, \ldots, X_{2n-1}) \\ &= H(\{X_i\}), \end{aligned}$$

where the inequality follows because conditioning reduces entropy.

Therefore,

$$\boxed{H(\{Y_i\}) = H(\{X_i\}), \qquad H(\{Z_i\}) = 2H(\{X_i\}), \qquad H(\{V_i\}) \geq H(\{X_i\}).}$$